

ONYX Healthcare Network

Where Every Life Matters.



THE TEAM!



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Introduction & Context

About ONYX



Introduction: Understanding ONYX Health Network

- **Overview:** This analysis explores patient records from **10 major hospitals across the U.S.**, offering deep insights into healthcare trends, costs, and patient demographics.
- **Dataset:** Includes over **55,000 unique patient records**, covering details like age, gender, medical conditions, billing information, and insurance providers.
- **Objective:** To uncover patterns in **patient demographics, medical conditions, treatment costs**, and the role of **insurance and hospitals** in shaping patient outcomes.
- **Goal:** Support **data-driven decisions** to improve Healthcare delivery, reduce costs, and enhance overall **patient experience and outcomes** across the network.

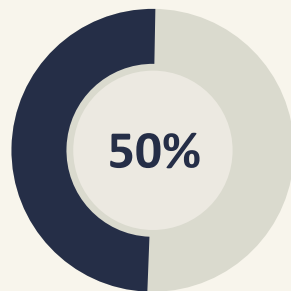


02

Who are our Patients?

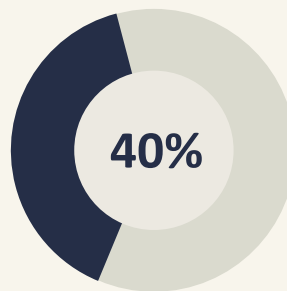
Demographic Overview

Gender Distribution



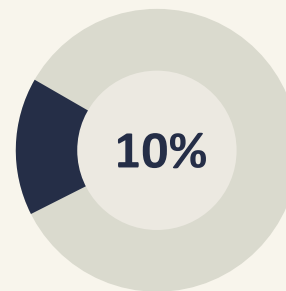
Females

27750 Females



Males

22200 Males



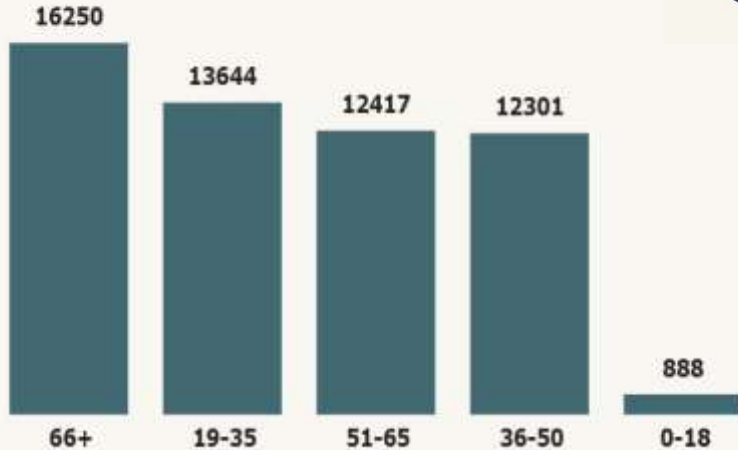
Non-Binary

5550 Non-Binary

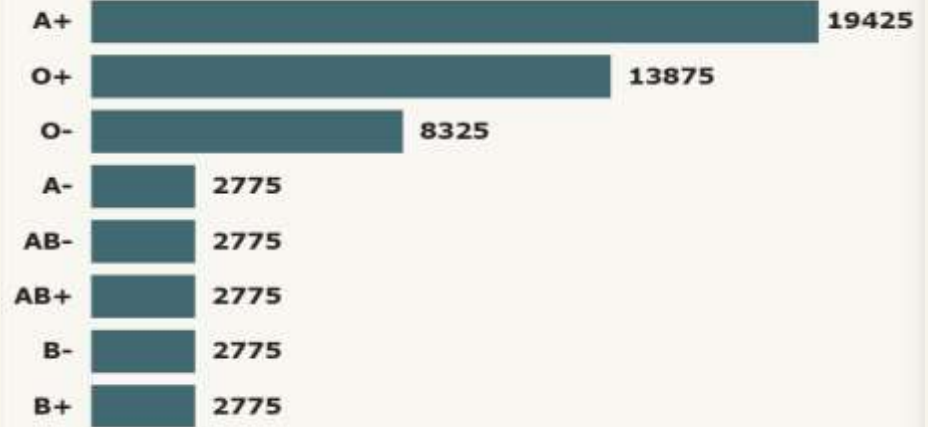


Age and Blood Type Distribution

Age Distribution



Blood Type Distribution



- majority of patients fall within the elderly category. Females made up a slightly larger share of total patients, and blood type A+ was the most common among all patients.

03

Medical Conditions

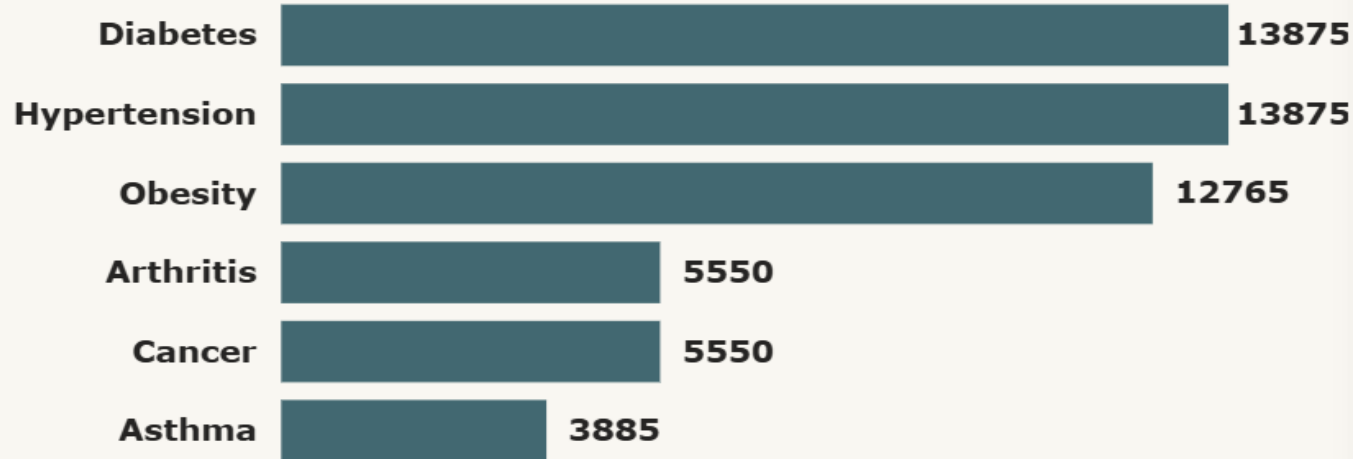
What Conditions Are We Dealing With?

“Chronic illnesses are our biggest challenge.”



Medical Conditions

Diagnosed Conditions

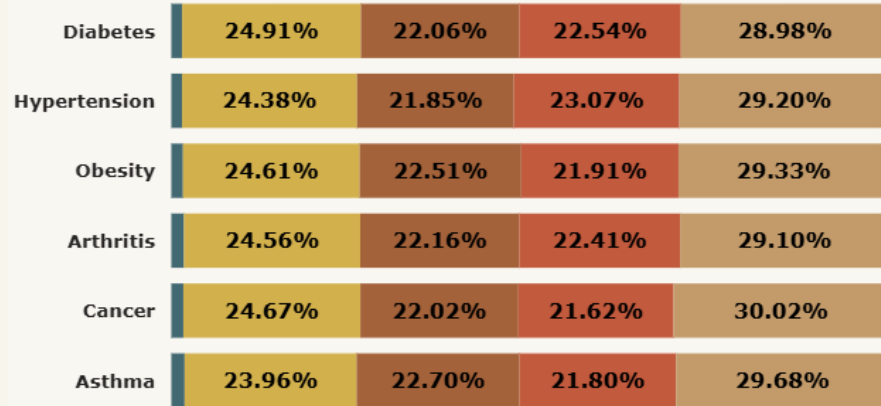


- The diagnosed medical conditions across the ONYX Health Network include **hypertension, diabetes, obesity, arthritis, asthma, and cancer.**

Medical Conditions

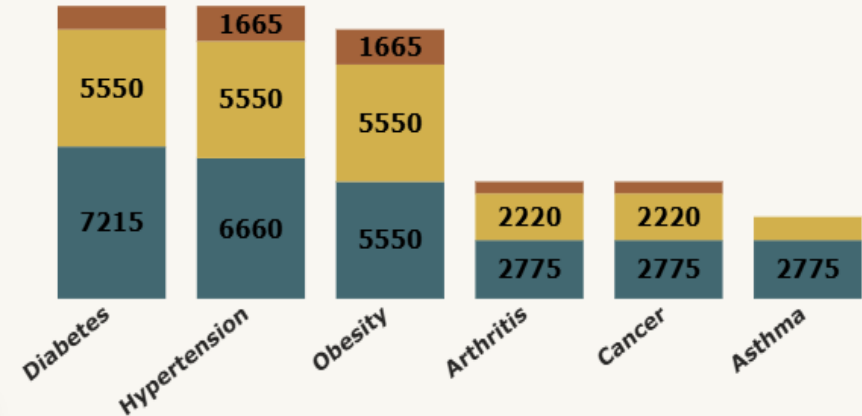
Medical Condition by Age Group

● 0-18 ● 19-35 ● 36-50 ● 51-65 ● 66+



Medical Condition by Gender

● Female ● Male ● Non-binary



- The conditions are more prevalent among **older adults**. Gender also plays a role, with all conditions prevalent in **female patients**.

Medical Conditions

Avg Billing Amount by Medical Condition



- Conditions like **asthma (\$25,678)** and **obesity (\$25,606)** are associated with a **slightly higher average billing**, indicating they may require more complex or extended treatment plans.

04

What's Driving Treatment Costs?

"Cost is inconsistent, and patients feel it."



Average Billing by Medical Condition



	Avg Billing(\$)
Asthma	25,678.33
Obesity	25,606.81
Hypertension	25,585.03
Cancer	25,503.80
Diabetes	25,503.26
Arthritis	25,298.15

- Chronic respiratory and lifestyle-related conditions like **Asthma**, **Obesity**, and **Hypertension** tend to generate higher average bills. These may reflect recurring treatment needs or specialized care requirements.



Average Billing by Hospital

	Avg Billing(\$)
Cedar Sinai Medical Center	25,728.81
Mayo Clinic	25,673.48
Massachusetts General Hospital	25,669.13
UCLA Medical Center	25,629.30
Johns Hopkins Hospital	25,577.61
Cleveland Clinic	25,536.14
Houston Methodist Hospital	25,535.65
UCSF Medical Center	25,346.41
Northwestern Memorial Hospital	25,333.51
New York Presbyterian Hospital	25,077.30



05

Does Insurance Coverage Matter?

"Insurance determines access."



Average Billing by Insurance Companies



Insurance Provider	Avg Billing(\$)
Cigna	25,727.04
United HealthCare	25,588.76
Medicare	25,494.43
Aetna	25,427.70



- **Cigna and United HealthCare** are associated with **higher billing amounts**, possibly due to more comprehensive plans or better coverage of higher-cost services.



☑ Insurance Coverage Matters

Insurance coverage plays a significant role in U.S. healthcare costs.

Insights from the ONYX dataset reveal clear differences in average billing amounts based on insurance provider:

- **Billing amounts vary** due to **negotiated rates** between insurers and hospitals.
- **Private and public insurers** (e.g., Cigna, United Healthcare, Medicare) show only **minor differences** in average billing across the ONYX dataset and while private insurance is often assumed to cost more, our data suggests that **billing levels remain broadly consistent** regardless of insurer type.
- Insurance coverage also impacts **access to care, length of stay, and treatment options**, all of which affect total costs.

RECOMMENDATIONS

Geriatric care programs

- Set up senior care programs to better serve ONYX's large aging population.

Chronic disease clinics

- Open chronic care clinics for diabetes, hypertension, and obesity, with education and follow-up to prevent serious health issues and cut long-term costs.

Billing & Cost Optimization

- Encourage performance-linked billing to align costs with health care quality and drive accountability.

Insurance enrollment support

- Launch in-hospital insurance assistance desks.
- Partner with insurers to simplify signups.
- Run awareness campaigns within hospitals and local communities.



CONCLUSION

- Chronic conditions like **asthma, obesity, and hypertension** have the highest average billing, highlighting a need for **better prevention and long-term care strategies**.
- Billing amounts are broadly consistent across hospitals, with only slight variations observed, even among high-profile institutions like Cedar Sinai and Mayo Clinic; suggesting minimal differences in cost structures or patient complexity.
- Private and public insurers show only minor differences in average billing, challenging the assumption that private insurance leads to significantly higher costs.
- These trends suggest opportunities to **optimize care delivery, standardize billing practices, and promote fair access to healthcare** through data-driven decisions and policy adjustments.



THANK YOU!





ONYX